

EX NO:05
21.02.2025

VIEWS AND INDEX

AIM:

To create different ways of views and index methods using SQL commands.

VIEW:

A view is an object that gives the user, a logical view of data from an underlying table.

CREATING VIEW:

BASE TABLE:

```
SQL> select * from studetails;
```

SID	NAME	DEP	YEAR
101	Arun	IT	2
102	Viji	CSE	1
103	Ram	AI	2
104	Suraj	FT	3
105	Yokarajan	IT	2
106	Rose	CSE	1
107	Sudee	CSE	2

7 rows selected.

DIFFERENT WAYS TO CREATE VIEWS:

- Horizontal view
- Vertical view
- Row/Column subset view
- Views using group by clause
- Views using join clause

CREATING THE VIEW BY SELECTING THE ATTRIBUTES FROM ONE TABLE (VERTICAL VIEW):

```
SQL> create view viewex as select sid, name, dept from studetails;
```

View created.

```
SQL> select * from tab;
```

TNAME	TABTYPE	CLUSTERID
VIEWEX	VIEW	

```
SQL> select * from viewex;
```

SID	NAME	DEP
101	Arun	IT
102	Viji	CSE
103	Ram	AI
104	Suraj	FT
105	Yokarajan	IT
106	Rose	CSE
107	Sudee	CSE

7 rows selected.

CREATING THE VIEW BY SELECTING THE ATTRIBUTES FROM ONE TABLE WITH PREDICATE(HORIZONTAL VIEW):

```
SQL> create view v1 as select * from studetails where name like '%e';
```

View created.

```
SQL> select * from v1;
```

SID	NAME	DEP	YEAR
106	Rose	CSE	1
107	Sudee	CSE	2

CREATING THE VIEW USING JOIN CLAUSE:

```
SQL> create view v as select s.sid, s.name, a.sem, a.cgpa from studetails  
s, acad a where s.sid = a.sid;
```

View created.

```
SQL> select * from v;
```

SID	NAME	SEM	CGPA
101	Arun	2	9.2
102	Viji	1	8.7
103	Ram	2	8.9
104	Suraj	3	8.1
105	Yokarajan	2	7.8
106	Rose	1	7.2
107	Sudee	2	9.3

7 rows selected.

CREATING THE VIEW USING GROUP BY CLAUSE:

```
SQL> create view v2 as select dept from studetails group by dept;
```

View created.

```
SQL> select * from v2;
```

DEP

AI

FT

CSE

IT

CREATING VIEW WITH SUB QUERY:

```
SQL> create view v3 as select name, dept from studetails where dept in  
(select dept from studetails group by dept);
```

View created.

```
SQL> select * from v3;
```

NAME	DEP
------	-----

Arun	IT
------	----

Viji	CSE
Ram	AI
Suraj	FT
Yokarajan	IT
Rose	CSE
Sudee	CSE

7 rows selected.

ROW/COLUMN SUBSET VIEW:

```
SQL> create view aa as select sid, name from studetails where dept in ('IT', 'CSE');
```

View created.

```
SQL> select * from aa;
```

SID NAME

101 Arun

102 Viji

180 Yokarajan

106 Rose

107 Sudee

110 Surya

6 rows selected.

CREATING THE VIEW USING AGGREGATE FUNCTION:

```
SQL> create view v5 as select max(year) as max_year_of_studetails from studetails;
```

View created.

```
SQL> select * from v5;
```

MAX_YEAR_OF_STUDETAILS

CREATING THE MATERIALIZED VIEW:

Certain database systems allow view relations to be stored, but they make sure that, if the actual relations used in the view definition change, the view is kept up-to-date.

Such views are called materialized views.

BASE TABLE:

```
SQL> select * from studetails;
```

SID	NAME	DEP	YEAR
101	Arun	IT	2
102	Viji	CSE	1
103	Ram	AI	2
104	Suraj	FT	3
105	Yokarajan	IT	2
106	Rose	CSE	1
107	Sudee	CSE	2

7 rows selected.

```
SQL> create view details as select sid, name, dept from studetails;
```

View created.

```
SQL> select * from details;
```

SID	NAME	DEP
101	Arun	IT
102	Viji	CSE
103	Ram	AI
104	Suraj	FT
105	Yokarajan	IT
106	Rose	CSE
107	Sudee	CSE

7 rows selected.

INSERTING RECORD IN BASE TABLE AND VIEWING IN THE VIEW TABLE:

```
SQL> insert into studetails values (108,'Sath', 'EEE', 4);  
1 row created.
```

```
SQL> select * from details;
```

```
select * from details;  
      SID NAME      DEP  
-----  
      108 Sath      EEE  
      101 Arun      IT  
      102 Viji      CSE  
      103 Ram       AI  
      104 Suraj     FT  
      105 Yokarajan IT  
      106 Rose      CSE  
      107 Sudee     CSE
```

```
8 rows selected.
```

DELETING RECORD IN BASE TABLE AND VIEWING VIEW TABLE:

```
SQL> delete from studetails where dept = 'FT';
```

```
1 row deleted.
```

```
SQL> select * from details;
```

```
      SID NAME      DEP  
-----  
      108 Sath      EEE  
      101 Arun      IT  
      102 Viji      CSE  
      103 Ram       AI  
      105 Yokarajan IT  
      106 Rose      CSE  
      107 Sudee     CSE
```

```
7 rows selected.
```

UPDATING IN BASE TABLE AND VIEWING VIEW TABLE:

```
SQL> update studetails set dept='IT' where name='Viji';
```

```
1 row updated.
```

```
SQL> select * from details;
```

SID	NAME	DEP
108	Sath	EEE
101	Arun	IT
102	Viji	IT
103	Ram	AI
105	Yokarajan	IT
106	Rose	CSE
107	Sudee	CSE

7 rows selected.

Updatable view:

The views that allows modification operations like INSERT, UPDATE, or DELETE to be performed on the underlying tables through the view.

For the view to be updatable:

- The from clause has only one database relation.
- The select clause contains only attribute names of the relation and does not have any expressions, aggregates, or distinct specification.
- Any attribute not listed in the select clause can be set to null; that is, it does not have a not null constraint and is not part of a primary key.
- The query does not have a group by or having clause.

BEFORE INSERTING VALUES IN VIEW TABLE:

```
SQL> select * from details;
```

SID	NAME	DEP
108	Sath	EEE
101	Arun	IT
102	Viji	IT
103	Ram	AI
105	Yokarajan	IT

106	Rose	CSE
107	Sudee	CSE

AFTER INSERTING THE RECORDS IN VIEW TABLE AND SEEING IN BASE TABLE:

SQL> insert into details values (109, 'Ajith', 'CSD');

1 row created.

SQL> insert into details values (110, 'Surya', 'IT');

1 row created.

SQL> select * from studetails;

SID	NAME	DEP	YEAR
108	Sath	EEE	4
101	Arun	IT	2
109	Ajith	CSD	
102	Viji	IT	1
103	Ram	AI	2
105	Yokarajan	IT	2
106	Rose	CSE	1
107	Sudee	CSE	2
110	Surya	IT	

9 rows selected.

DELETING IN VIEW TABLE AND VIEWING BASE TABLE:

SQL> delete from details where name like '__th';

1 row deleted.

SQL> select * from studetails;

SID	NAME	DEP	YEAR
101	Arun	IT	2
109	Ajith	CSD	
102	Viji	IT	1
103	Ram	AI	2
105	Yokarajan	IT	2
106	Rose	CSE	1
107	Sudee	CSE	2
110	Surya	IT	

8 rows selected.

UPDATING IN VIEW TABLE AND VIEWING BASE TABLE:

SQL> update details set sid = 180 where name in (select name from studetails where name = 'Yokarajan');
1 row updated.

SQL> select * from studetails;

SID	NAME	DEP	YEAR
101	Arun	IT	2
109	Ajith	CSD	
102	Viji	IT	1
103	Ram	AI	2
180	Yokarajan	IT	2
106	Rose	CSE	1
107	Sudee	CSE	2
110	Surya	IT	

8 rows selected.

DROPPING VIEW:

SQL> create view v5 as select max(year) as max_year_of_studetails from studetails;

View created.

SQL> select * from v5;

MAX_YEAR_OF_STUDETAILS

2

SQL> drop view v5;

View dropped.

INDEX:

Index is used to access the records faster.

It occupies some space.

TABLE:

SQL> select * from courses;

CID	CNAME
101	Maths
103	DBMS

107 WT

105 OS

104 DAA

CREATING INDEX:

SINGLE INDEX:

```
SQL> create index i1 on courses(cname);
```

Index created.

COMPOSITE INDEX:

```
create index i2 on courses(cname, cid);
```

Index created.

DROP INDEX:

```
SQL> drop index i1;
```

Index dropped.

CREATE TABLE EXTENSION:

BASE TABLES:

```
SQL> select * from personal;
```

SID	NAME	AGE	ADDRE
1	arun	19	Salem
2	viji	18	Cbe
3	ram	20	Erode
4	suraj	21	Cbe
5	yokarajan	19	Karur
6	rose	18	erode
7	sudee	20	Salem
8	sath	22	Karur

8 rows selected.

```
SQL> select * from academic;
```

SID	DEPT	YEAR
1	it	2
2	cse	1
3	ai	2
4	ft	3
5	it	2
6	cse	1
7	cse	2
8	eee	4

8 rows selected.

CREATE TABLE EXTENSION:

USING JOIN:

```
SQL> create table det as select p.sid, p.name, p.age, a.dept, a.year from
personal p, academic a where p.sid = a.sid;
```

Table created.

```
SQL> select * from tab;
```

TNAME	TABTYPE	CLUSTERID
DET	TABLE	

```
SQL> select * from det;
```

SID	NAME	AGE	DEPT	YEAR
1	arun	19	it	2
2	viji	18	cse	1
3	ram	20	ai	2
4	suraj	21	ft	3
5	yokarajan	19	it	2
6	rose	18	cse	1
7	sudee	20	cse	2
8	sath	22	eee	4

UPDATING IN THE BASE TABLE:

SQL> update personal set name='surya' where sid=4;

1 row updated.

SQL> update academic set dept='it' where sid=3;

1 row updated.

AFTER UPDATING BASE TABLE RETERVING THE RECORDS FROM THE CTE TABLE:

(THE RECORDS ARE NOT UPDATED)

SQL> select * from det;

SID	NAME	AGE	DEPT	YEAR
1	arun	19	it	2
2	viji	18	cse	1
3	ram	20	ai	2
4	suraj	21	ft	3
5	yokarajan	19	it	2
6	rose	18	cse	1
7	sudee	20	cse	2
8	sath	22	eee	4

8 rows selected.

WITH OUT USING WHERE CLAUSE:

SQL> create table exe as select name, age, sid from personal;

Table created.

SQL> select * from exe;

NAME	AGE	SID
arun	19	1
viji	18	2
ram	20	3

surya	21	4
yokarajan	19	5
rose	18	6
sudee	20	7
sath	22	8

8 rows selected.

USING WHERE CLAUSE:

SQL> create table exe1 as select name, age from personal where name not like '%e' and name not like '%i';

Table created.

SQL> select * from exe1;

NAME	AGE
-----	-----
arun	19
ram	20
surya	21
yokarajan	19
sath	22

DIFFERENCE BETWEEN AND UPDATABLE VIWES:

S.NO	CREATE TABLE EXTENSION	VIEW
01	CTE occupies memory	View does not occupy memory
02	CTE always reflects the records in the base table only at the time of creation. Updates in the base table are not reflected in the CTE table.	Views are updated whenever the base table is updated
03	CTE is a physical table	View is a virtual table

CONTENTS	MARKS ALLOTED	MARKS OBTAINED
Aim, Algorithm, SQL, PL/SQL	30	
Execution and Result	20	
Viva	10	
Total	60	

RESULT:

Thus creation of different views using SQL commands is completed